CHILAKALA SURYA PRAKASH

Java Backend Developer | Spring Boot · Microservices · AWS

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PROFESSIONAL SUMMARY

Java Backend Developer with hands-on experience building RESTful microservices and distributed systems using **Java 17/21, Spring Boot, Apache Kafka, Redis, and AWS**. Experienced in designing ACID-compliant transactions, stateless JWT security, real-time event streaming, and automated CI/CD pipelines with GitHub Actions and Docker. Strong foundation in **OOP, Collections Framework, Streams API**, and testing with JUnit 5 and Mockito.

TECHNICAL SKILLS

Languages & Core Concepts:	Java (17/21), Python, SQL, OOP, Collections Framework, Streams API, Exception Handling
Frameworks & Backend:	Spring Boot 3.x, Spring Data JPA (Hibernate), Spring Security, RESTful APIs, Microservices
Messaging & Caching:	Apache Kafka (Event Streaming), Redis (Idempotency & Distributed Locking)
Databases & Storage:	PostgreSQL, MySQL, H2 In-Memory, ACID Compliance, Custom JPQL Queries
Cloud & DevOps:	AWS (EC2, VPC, IAM), Docker, Docker Compose, GitHub Actions (CI/CD), Maven
AI Integration & Testing:	Google Gemini 2.5 API (Zero-Shot Prompting), JUnit 5, Mockito, Postman, Git/GitHub

WORK EXPERIENCE

Data Analytics & Visualization Simulation — Accenture North America (Forage) May 2024 – Jul 2024

Remote

- Completed an analytics job simulation advising a business client on content performance trends across 7 datasets.
- Cleaned, modeled, and analyzed data using exploratory analysis techniques to uncover audience engagement patterns.
- Delivered an executive summary deck and video walkthrough communicating actionable insights to client stakeholders.

PROJECTS

Digital Wallet Microservice · Java 17 · Spring Boot · Apache Kafka · Redis · PostgreSQL · Docker · AWS GitHub ↗

- Built an event-driven ledger microservice in **Java 17 and Spring Boot 3.2** for peer-to-peer money transfers, using `@Transactional` to guarantee **ACID integrity** and `BigDecimal` for precision arithmetic.
- Implemented an idempotency filter with **Redis atomic locks (SETNX, 24h TTL)** to block duplicate transfer requests and prevent double-debiting during client retries (returning HTTP 409 Conflict).
- Streamed transaction audit records asynchronously to **Apache Kafka** (`transaction-events`), masking account numbers (`****1001`) to safeguard sensitive customer data.
- Automated CI/CD using **GitHub Actions** to run unit, slice (`@WebMvcTest`, `@DataJpaTest`), and integration tests on commit; deployed the containerized stack to **AWS EC2** with configured VPC Security Groups and Linux swap memory.

AI-Powered Expense Tracker & Budget Guardrail API · Java 21 · Spring Boot · Google Gemini AI · JWT · JPA GitHub ↗

- Developed a personal finance REST API in **Java 21 and Spring Boot 3.4**, integrating **Google Gemini 2.5 Flash** to parse natural language expense notes (e.g., "Spent $40 on fuel") into structured database entities.
- Engineered a real-time budget guardrail engine that calculates cumulative month-to-date spending on every transaction write, returning instant status alerts (`SAFE`, `WARNING`, `EXCEEDED`).
- Secured endpoints using **Spring Security filter chains and JWT tokens (HMAC-SHA256)** with BCrypt password encryption to ensure isolated multi-tenant user access.
- Wrote optimized **JPQL aggregation queries (`SUM`, `GROUP BY`)** to compute monthly category breakdowns directly in the database, reducing server memory usage and API latency.

Mental Health Risk Prediction Pipeline · Python · scikit-learn · XGBoost · K-Means GitHub ↗

- Engineered a machine learning pipeline in Python on psychological survey data, computing normalized depression, anxiety, and stress risk scores across 10 personality traits.
- Applied **K-Means clustering** (evaluated via Elbow and Silhouette methods, *k=3*) to categorize respondents into 3 clinical severity tiers (*Mild, Moderate, Severe*) as target labels.
- Trained and benchmarked 4 algorithms (**XGBoost, Random Forest, MLP, Logistic Regression**) using `RandomizedSearchCV`; **XGBoost** delivered best results with **98% accuracy and weighted F1-score**.

EDUCATION

Bachelor of Engineering — Computer Science & Engineering July 2025

Gopalan College of Engineering and Management, Bangalore **CGPA: 7.1 / 10**

CERTIFICATIONS

- 100 Days of Code: Complete Python Bootcamp — Udemy
- Data Visualization with Power BI — Great Learning Academy
- Data Analysis with Python — freeCodeCamp.org
- AWS Certified Developer - Associate — Infosys Springboard